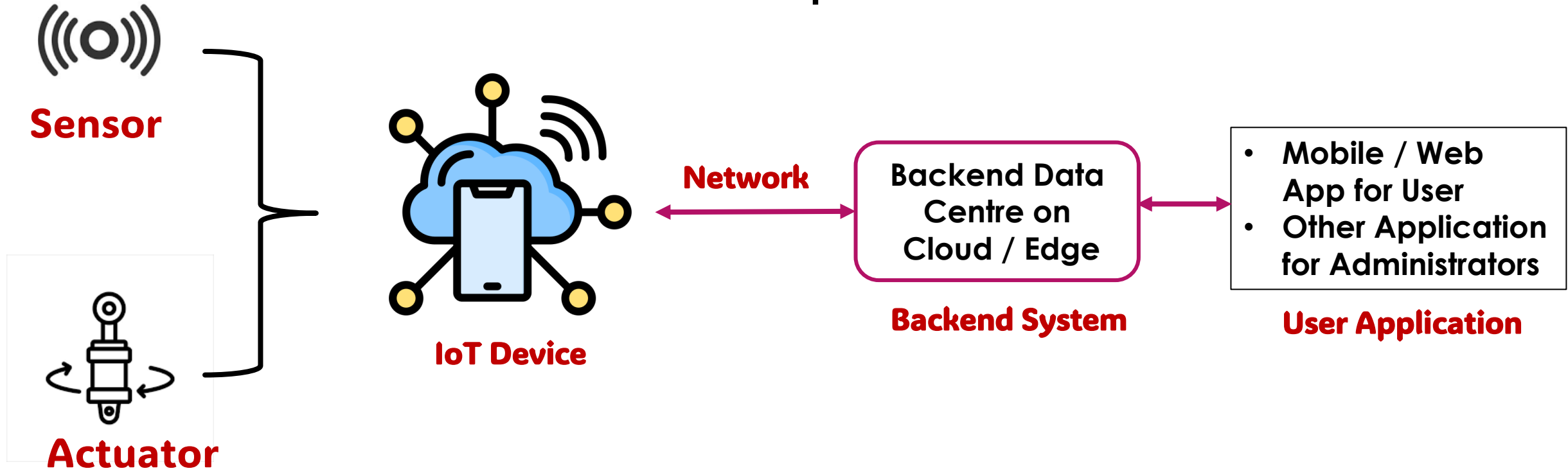
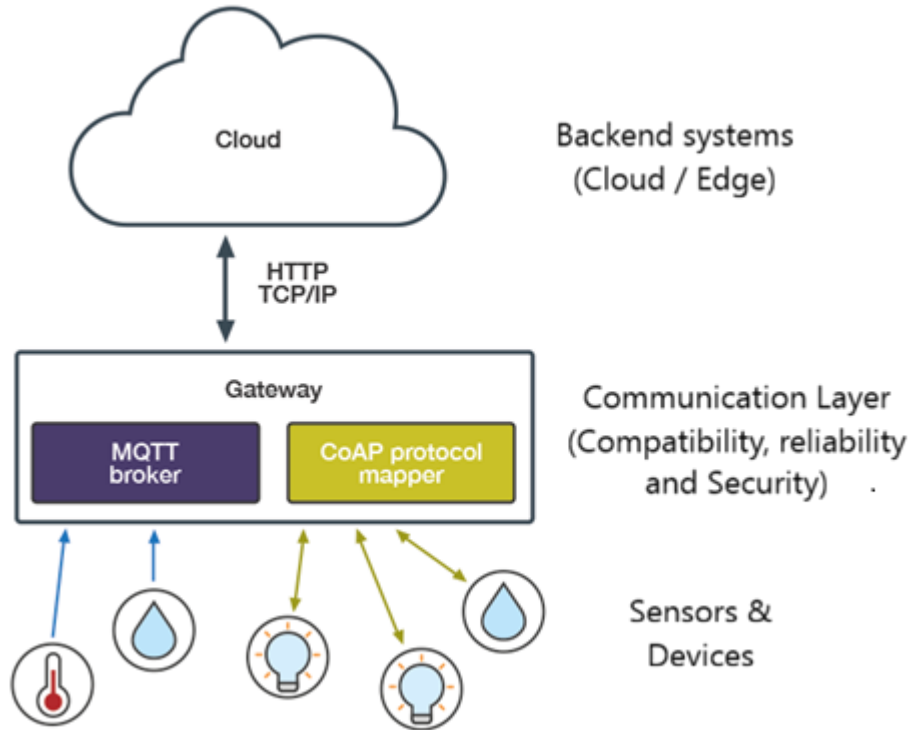


IoT Sensor and Actuators

IoT Architecture Recap



IoT Sensor and Device Integration



IoT Sensor and Device Integration

- Connects physical sensors (temperature, motion, etc.) with software, applications, and cloud platforms.
- **Key Components**
 - **Sensors** - The "eyes and ears" gathering data (temperature, humidity, light, etc.).
 - **Connectivity** - Protocols (MQTT, HTTP) and networks (Wi-Fi, cellular) that link devices.
 - **Gateways/Platforms** - Central hubs for data aggregation and initial processing (edge/cloud).
 - **Applications/Analytics** - Software that processes data, generates insights, and enables actions.

Key Factors

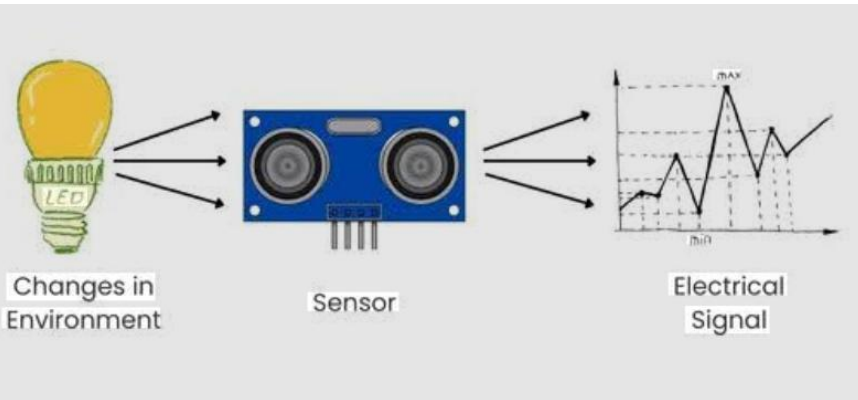
Right sensors
temperature, GPS,
RFID, LiDAR, etc.

Communication protocols
(Wi-Fi, LoRaWAN, BLE, MQTT).

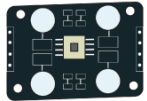
Reliable Connections
using Wi-Fi, Bluetooth,
LoRa, NB-IoT, or 5G.

Secure Data Transmission
(MQTT, CoAP, HTTP).

What is a Sensor ?



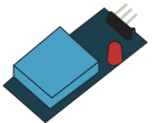
Metal Sensor



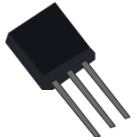
Color Sensor



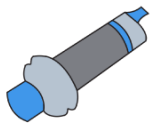
Light Sensor



Humidity Sensor



Temperature Sensor



Proximity Sensor

Senses Real World data

- Transforms Physical World Data into electrical signal (Analog)
- Made of Materials sensitive to physical / chemical change → change in voltage, current, resistance or charge

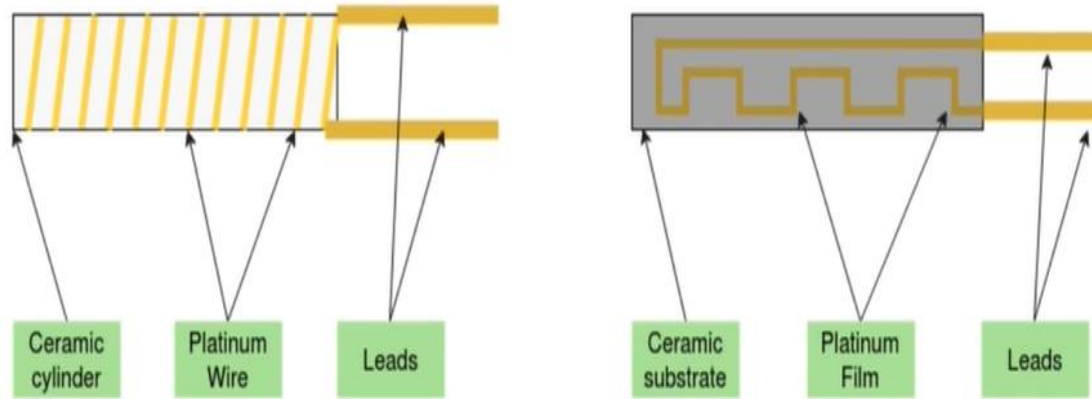
Analog to Digital Conversion

- **Sampling Rate** – Used to convert analogue signal into digital signal
- **Nyquist criterion** – measures signal fidelity and accuracy – [Link](#)
- **Resolution** - Set of digital values (steps) – represent in number of bits
- **Multiple approaches** - Complex Algorithms are used for fine tuning

Digital Processing

- **Display locally** - (DRO-Digital Read out), ex. weighing machine
- **Store Locally** - ex. last ten body temperature readings
- **Transmit up stream** - ex. smoke alarm info to the fire station

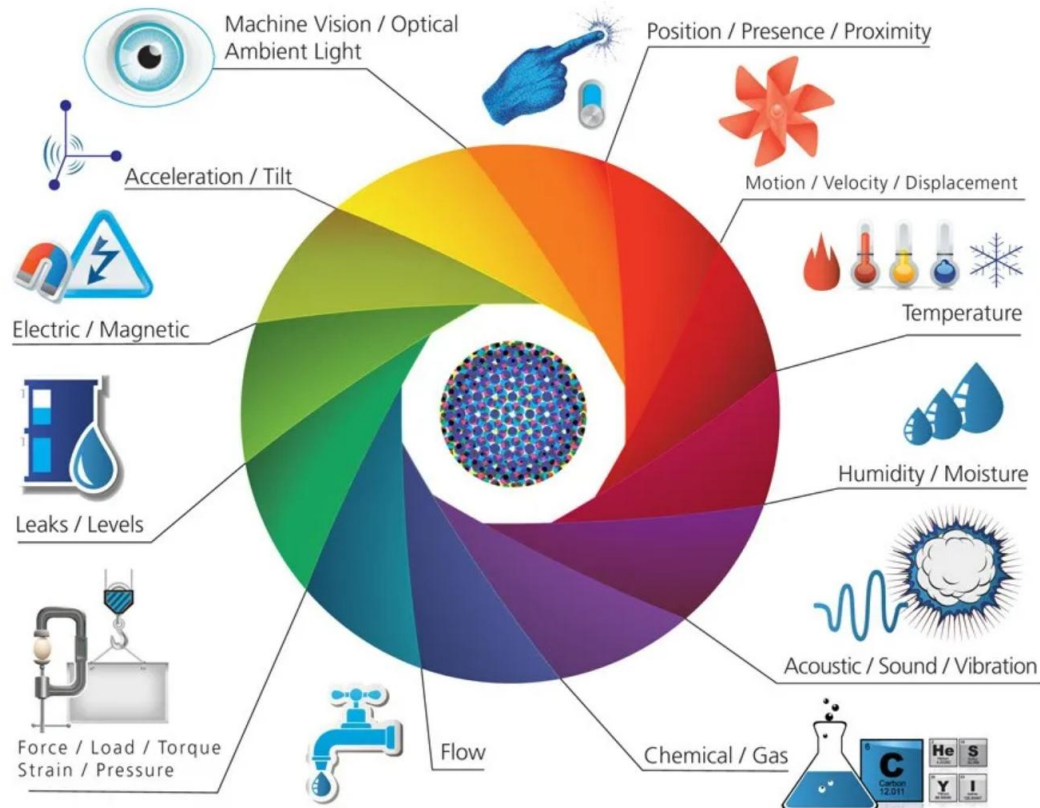
What is a Sensor ?



PT100 – Temperature Sensor

- **Made of Platinum** – Robust metal - Resistant to corrosion and chemicals and stable at high temperatures
- **Linear resistance change** based on temperature – 100 ohms resistance at 0 °C - there after steady change in resistance with change in temperature.
- Less material used so **cost-effective**
- **Construction** : Platinum wire wrapped around an insulated core OR platinum thin-film embedded in a ceramic substrate

What is a Sensor ?



Environmental

- **Temperature** - Thermistor, Thermocouple
- **Humidity** - Psychrometric, Chilled Mirror Hygrometers
- **Pressure** - Piezo resistive Strain Gauge, MEMS Barometric Sensor

Entities

- **Location** – GPS
- **Movement** - Accelerometer, Gyroscope, Tachometer

Particle Density

- **Air quality** (CO, NO, SO₂, O₃, Particulates, CO₂, etc.)
- Infrared, Electrochemical
- Smoke, Gas - Ionization, Photoelectric, Infrared

What is a Sensor ?



Everything is getting smarter – Smart Devices, Smart Homes, Smart Cities Smart Health, Smart Transportation and Mobility

Over **10 billion IoT devices** worldwide, and this number is expected to reach **30 billion by 2030**

Advancement in **communication technologies like 5G** and the **upcoming 6G**, combined with improvements in **computing power** and **device miniaturization** make it possible to deploy sensors everywhere.

Fleet
management

Micromobility

Building
Monitoring

Environmental
Monitoring

Predictive
Maintenance

Smart Farming
& Watering

Medical IoT

What is a Actuator ?

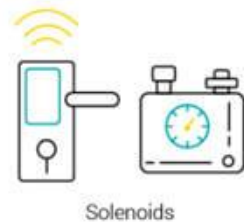
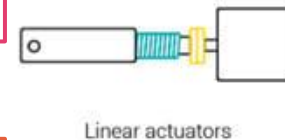
An IoT Actuator is a component that converts electrical signals from an IoT system into physical actions or movements

Responsible for effecting a change

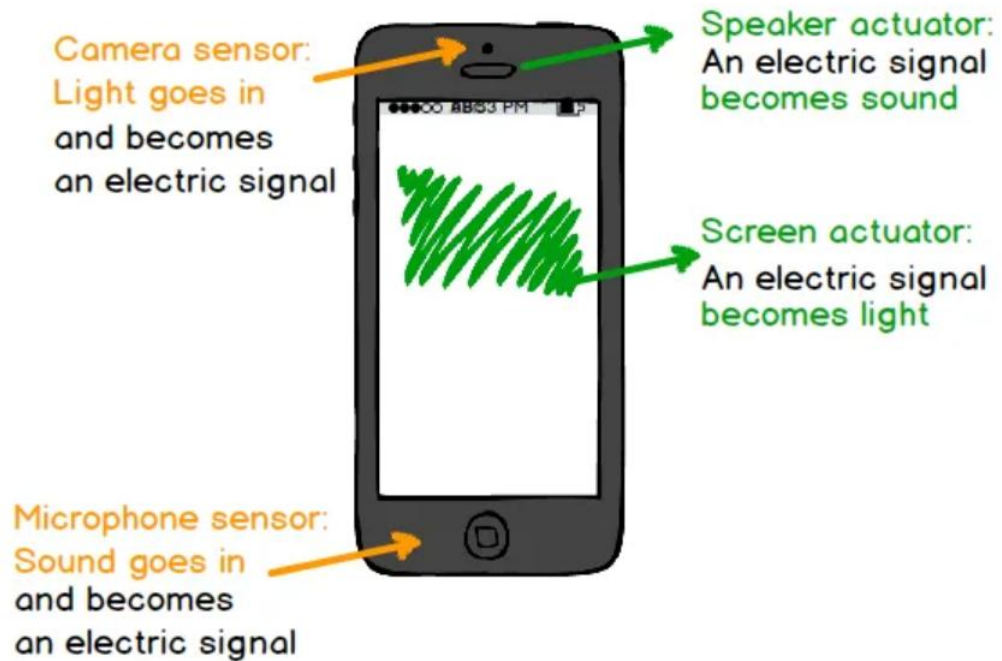
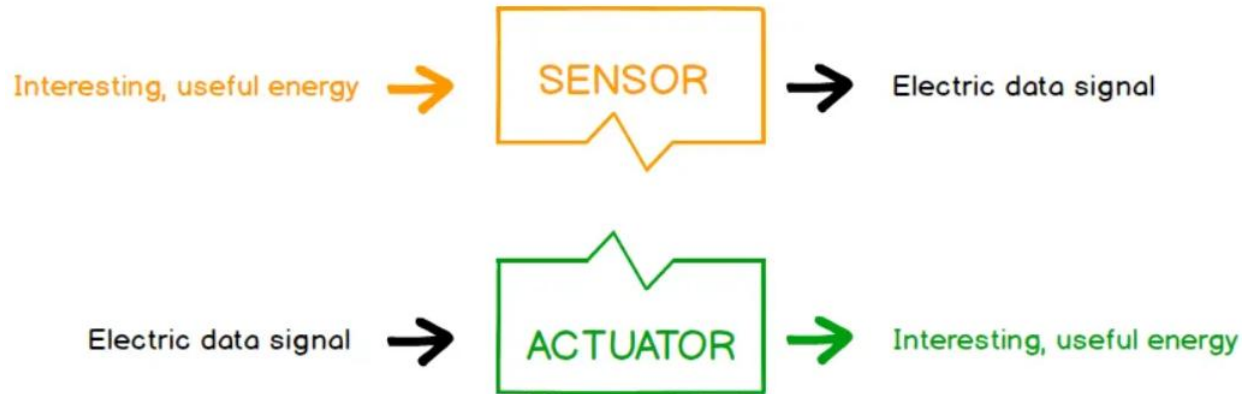
- Physical movement causing a changing E.g. - starting pump when water level is low
- **General Types:** Hydraulic , Pneumatic, Mechanical, Electrical, Magnetic, Thermal

IoT: Change triggered by electrical signal

- **Sound** - Electroacoustic Transducer, Solenoid movement
- **Switching** - Relays, Electromechanical
- **Movement** - Stepper motor, Servomotor, Solenoid



What is a Actuator ?



Sensor

- Microphones
- Camera

Actuator

- Speakers
- Screen